Financial RAG — Noun-Phrase Collision Elimination + Chain-of-Verification
Flowchart: One Query's Path Through the Online Query Plane

LEGEND
Deterministic step no model involved
LLM call — generative numbered = one of the six permitted call sites
Embedding / cross-encoder — non-generative model
Mixed: deterministic + embedding model
Decision point
User submits a question
Query Analyzer noun-phrase extraction + collision detection (lexicon-only)
NPCE enabled for this run?
off — C1 / C2
on — C3 / C4
Query Splitter **CALL SITE 3 — LLM** (slot parsing only) + deterministic Cartesian product (the split itself is code, not the model)
Single pass-through sub-query — no split, identity transform
Query-side Placeholder Remap each resulting sub-query names exactly one entity, one metric, one period
Fan out: N sub-queries (N = 1 if NPCE off)
Per sub-query s_i — repeated independently for each of the N sub-queries
Metadata Filter SQL over Postgres — narrows candidates by this sub-query's entity/period/metric tags
Hybrid Retrieve BM25 (det) + Dense (embedding)
Rank Fusion + time-aware weighting
Cross-encoder Rerank non-generative model
FFF Rewriter **CALL SITE 4 — LLM** rewrites top propositions into Fully-Formatted Facts
Numeric-verbatim guard: does every digit & period trace verbatim to the source proposition?
passes
fails
Fact-set / Query Pairer keyed binding only
Generator **CALL SITE 5 — LLM** answers from this sub-query's paired FFFs only
Reject this FFF — metric honestly reported 'not provided'
Sub-answer a_i
Draft Assembler DET — collects all N sub-answers, introduces no new facts → draft D
CoVe enabled for this run?
off — C1 / C3
on — C2 / C4
Extract atomic claims from draft D **CALL SITE 6a — LLM**
Template one verification question per claim DET — deterministic construction
Execute verification independently **CALL SITE 6b — LLM** draft D is structurally absent from this call's context — not a prompt instruction
Claim type?
numeric
prose
Deterministic numeric comparison DET — value equality, never an LLM judgment
Prose comparison **CALL SITE 6c — LLM**
Draft D passes straight through, unverified
Per-claim verdict: match / conflict / unsupported
Abstention Gate DET — rules over verification results (all-or-nothing across a question's claims)
Every required claim verified & evidence-supported?
no
yes
Abstain — 'not found in source', no citation, no guess
Placeholder Reverse-Mapper DET — restores real entity names
Citation Binder DET — looks up proposition_id metadata **an LLM never writes a citation**
Final answer + citations + verification trace